

BALASUBRAMANI B

Software Engineer | Full Stack Developer | Backend Developer

Bengaluru, India | +91 7639982093 | bbalasubramani.dev@gmail.com | [linkedin.com/in/balasubramani-dev](https://www.linkedin.com/in/balasubramani-dev) | github.com/bbalasubramani

Summary

CSE 2026 graduate with 3 production-deployed full-stack apps (React + Node.js + PostgreSQL) and Gemini API integration in a live labor marketplace. Built JWT-secured REST APIs with bcrypt auth, automated CI/CD via GitHub Actions, and a cross-runtime Node.js/Python prediction API — all independently shipped and running in production.

Skills

Languages: Python, JavaScript, SQL

Frontend: React.js, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs, JWT, bcrypt, Gemini API, Firebase Auth

Databases: PostgreSQL, MongoDB, Firebase Firestore, Supabase

DevOps: Git, GitHub Actions (CI/CD), Vercel, Render, Netlify, Postman

Experience

Android Developer Intern

February 2026 – May 2026

MindMatrix Technologies

Remote

- Integrated Gemini REST API into a live Android marketplace app — built JSON request/response handling, image-to-Base64 encoding for vision calls, HTTP error handling, and prompt engineering for 3 product features: natural-language search classification, skill description generation, and badge suggestions.
- Designed a centralized FirebaseRepository layer (Auth, Firestore, Storage) handling OTP and email authentication, NoSQL document CRUD with ownership-scoped writes, real-time snapshot listeners for live worker availability, and Firestore security rules with default-deny access control; implemented Haversine distance calculation for proximity-based worker ranking.

Projects

Job Tracker App | *React, Node.js, Express, PostgreSQL, JWT, GitHub Actions* | job-tracker-frontend-red.vercel.app | [GitHub](#) 2026

- Designed a normalized PostgreSQL schema (users, jobs, applications tables) with foreign key constraints verified across 8 registered test users; secured all 12 REST API endpoints with JWT and bcrypt authentication and parameterized queries to prevent SQL injection.
- Configured a multi-service cloud deployment — React on Vercel, Node.js/Express on Render, PostgreSQL on Supabase — with separate development and production environment variables; set up a GitHub Actions CI pipeline running ESLint and build checks on every push to prevent broken builds from reaching production.
- Built centralized input validation middleware in Express.js across all API endpoints to reject malformed requests before the database layer; achieved zero error-state bug reports over a 2-week production usage period.

Book Barter System | *React, Node.js, Express.js, MongoDB, REST APIs* | bookbartersystem.netlify.app | [GitHub](#) 2025

- Built a peer-to-peer book exchange platform on the MERN stack with JWT authentication and bcrypt password hashing; enforced role-based access control on all write endpoints using custom Express.js auth middleware.
- Implemented a barter request workflow (create, accept, reject) with ownership-scoped MongoDB queries; integrated OpenLibrary API returning 5 search results per query; auth middleware blocked 100% of unauthorized write attempts during penetration testing.
- Added a `/health` monitoring endpoint integrated with BetterStack for uptime alerting; configured Netlify and Render deployments with production-ready CORS settings.

BirthSense – Delivery Mode Predictor | *Node.js, Express.js, Python, scikit-learn, MongoDB* | birthsense.netlify.app | [GitHub](#) 2025

- Built a Node.js/Express REST API (14 endpoints) connecting to a Python Random Forest model trained on 5,842 patient records; managed cross-runtime communication between Node.js and Python to return delivery mode predictions (Normal vs. Caesarean) with confidence scores.
- Achieved 85%+ test accuracy on the Random Forest classifier; built the complete frontend in HTML, CSS, and JavaScript and configured full-cycle deployment across Netlify and Render with production environment setup.

Education

B.E. Computer Science Engineering

May 2026

HKBK College of Engineering

Bengaluru, Karnataka

- CGPA: **8.5 / 10**
- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, DBMS and SQL, Operating Systems, Computer Networks, Software Engineering

Certifications

- NPTEL – Cloud Computing | IIT Kharagpur (via SWAYAM), 2025
- NPTEL – Machine Learning | IIT Madras (via SWAYAM), 2024